SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

b) Let the input to Δ_m be ε while the output be δ. Then a calculation shows that $\varepsilon = -S\delta$ where $S = (I+GK)^{-1}$ is the sensitivity. Using the small gain theorem, to satisfy the robustness design requirement, it is sufficient that $\|\Delta_m(j\omega)S(j\omega)\| < 1, \forall\omega$. This can be satisfied if $\|W_m S\|_\infty < 1$, where $W_m = w_m I$.

A calculation shows that, when $\Delta_m = 0$,

$$u(s) = -K(s)S(s)d(s).$$

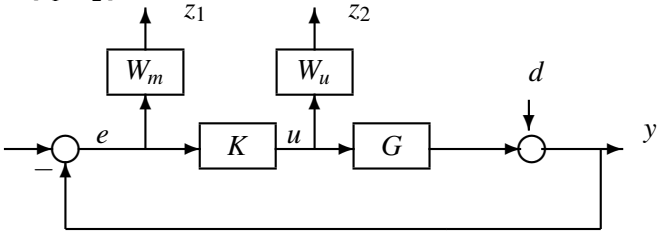
Thus

$$\|u(j\omega)\| \le \|K(j\omega)S(j\omega)\|\|d(j\omega)\|,$$

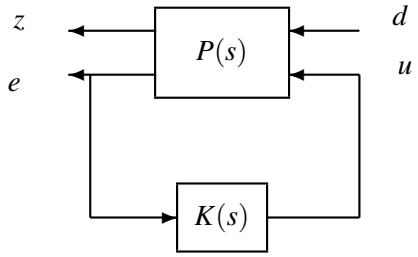
It follows that a sufficient condition to achieve the performance design specification is $\|K(j\omega)S(j\omega)\| < |w_u^{-1}(j\omega)|$ for all ω or equivalently $\|W_u KS\|_\infty < 1$, where $W_u = w_u I$.

To satisfy all design requirements, it is sufficient that $\left\| \begin{bmatrix} W_m S \\ W_u KS \end{bmatrix} \right\|_\infty < 1$.

c) So, the design specifications reduce to the requirement that the transfer matrix from d to $z = [z_1^T \;\; z_2^T]^T$ in the following diagram has $\mathscr{H}_\infty$-norm less than 1.



d) The generalized regulator problem: find an internally stabilizing K such that $\|T_{zd}\|_\infty < 1$



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|c} -W_m & -W_m G \\ 0 & W_u \\ \hline -I & -G \end{array}\right].$$

2. a) i) A is stable if $A^TP+PA \prec 0$ with $P \succ 0$. To prove this, let $Az=\lambda z$, $z \neq 0$. Then multiplying the inequality from the left by z' and from the right by z gives $(\lambda+\bar{\lambda})z'Pz<0$. Since $P \succ 0$ it follows that $z'Pz>0$ and it follows that $\lambda+\bar{\lambda}<0$ so that A is stable.

ii) Assume that $x(0)=0$. Then $\|H\|_\infty < \gamma$ if and only if $J := \int_0^\infty [y^Ty-\gamma^2u^Tu]dt<0$, for all $u(t)$ such that $\|u\|_2<\infty$. Since $\|u\|_2$ is bounded, $\lim_{t\to\infty} x(t)=0$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}[x^TPx]dt = x(\infty)^TPx(\infty)-x(0)^TPx(0)=0$. So,

$$0=\int_0^\infty \left(\dot{x}^TPx+x^TP\dot{x}\right)dt=\int_0^\infty [x^T(A^TP+PA)x+x^TPBu+u^TB^TPx]dt.$$

Using $y=Cx+Du$ and adding the last expression to J

$$\begin{aligned} J &= \int_0^\infty [x^T(A^TP+PA+C^TC)x+2x^T(PB+C^TD)u+u^T(D^TD-\gamma^2I)u]dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^TP+PA+C^TC & PB+C^TD \\ B^TP+D^TC & D^TD-\gamma^2I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J<0$ if $M \prec 0$. So A is stable and $\|H\|_\infty<\gamma$, if $M \prec 0$ and $P \succ 0$ since $M \prec 0$ also implies that $A^TP+PA \prec 0$.

iii) The dual condition is: A is stable and $\|H\|_\infty<\gamma$ if and only if A^T is stable and $\left\|H^T\right\|_\infty<\gamma$. Using Part a(ii) (replacing (A,B,C,D) by (A^T,C^T,B^T,D^T)), A is stable and $\|H\|_\infty<\gamma$ if there exists $P \succ 0$ such that

$$N := \begin{bmatrix} AP+PA^T+BB^T & PC^T+BD^T \\ CP+DB^T & DD^T-\gamma^2I \end{bmatrix} \prec 0.$$

b) i) The state equations from the diagram are:

$$\dot{x}=(A+BF)x+d, \qquad y=Cx.$$

Therefore $T_{yd}(s)=C\left(sI-(A+BF)\right)^{-1}$.

ii) Using the answer to Part a(iii), replacing A by $A+BF$, B by I and setting $D=0$, there exists a feasible matrix F if there exists $P \succ 0$ such that

$$\begin{bmatrix} (A+BF)P+P(A+BF)^T+I & PC^T \\ CP & -\gamma^2I \end{bmatrix} \prec 0.$$

Since the only nonlinearity is due to the product FP, we let $Z=FP$ and so there exists a feasible gain matrix F if there exists $P \succ 0$ and Z such that

$$\begin{bmatrix} AP+PA^T+BZ+Z^TB^T+I & PC^T \\ CP & -\gamma^2I \end{bmatrix} \prec 0.$$

3. a) Let $V = x'Px$ and set $u = -Fx$. Provided that $P = P' \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Px + x'P\dot{x} = x'\left(A'P + PA - F'B'P - PBF\right)x.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$\int_0^\infty x'\left(A'P + PA - F'B'P - PBF\right)xdt = -x_0'Px_0.$$

Using the definition of J, adding the last equation and completing a square:

$$J = x_0'Px_0 + \int_0^\infty \{x'[A'P + PA + I - PBB'P]x + \|(F - B'P)x\|^2\}dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = B'P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation,

$$A'P + PA + I - PBB'P = 0.$$

It follows that the minimum value of J is $x_0'Px_0$.

b) We need to prove that $A_c := A - BB'P$ is stable. The Riccati equation can be written as $A_c'P + PA_c + I + PBB'P = 0$. Let $\lambda \in \mathscr{C}$ be an eigenvalue of A_c and $z \neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be z' and z respectively gives $(\lambda + \bar{\lambda})z'Pz + z'z + z'PBB'Pz = 0$. Since $P \succ 0$ and $z \neq 0$, $z'Pz > 0$, $z'z > 0$ and $z'PBB'Pz \geq 0$. It follows that $\lambda + \bar{\lambda} < 0$ and the closed loop is stable.

c) The loop-gain is given by $L(s) = F(sI - A)^{-1}B$. Let λ be a zero of $L(s)$ so that,

$$\begin{bmatrix} A - \lambda I & B \\ F & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

for some $v_1,\ v_2$ with $v_1 \neq 0$ since B has full column rank. Then,

$$Av_1 = \lambda v_1 - Bv_2,$$

and,

$$Fv_1 = 0 \Rightarrow B'Pv_1 = 0.$$

Consider the Riccati equation. Then,

$$\begin{aligned} 0 &= v_1'[A'P + PA - PBB'P + I]v_1 \\ &= (\bar{\lambda}v_1' - v_2'B')Pv_1 + v_1'P(\lambda v_1 - Bv_2) + v_1'v_1 \\ &= (\bar{\lambda} + \lambda)\underbrace{v_1'Pv_1}_{>0} + \underbrace{v_1'v_1}_{>0} \end{aligned}$$

Hence $\lambda + \bar{\lambda} < 0$ and $L(s)$ is minimum-phase.

4. ai) A suitable Lyapunov function is $V = x^T P x$ where $P = P^T$. Provided that $P = P^T \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we have $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}^T P x + x^T P \dot{x} = x^T \left(A^T P + PA - F^T B^T P - PBF\right) x.$$

Hence $\int_0^\infty \dot{V}(t)dt = \int_0^\infty x^T \left(A^T P + PA - F^T B^T P - PBF\right) x dt = -x_0^T P x_0$.

aii) Adding the last equation to J_0 and completing a square:

$$J_0 = x_0^T P x_0 + \int_0^\infty \{x^T [A^T P + PA + C^T C - PBB^T P] x + \left\|(F + B^T P)x\right\|^2\} dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = -B^T P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation $A^T P + PA + C^T C - PBB^T P = 0$.

bi) Let $J = \|z\|_2^2 - \gamma^2 \|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'B'X + XBF\right)x + x'Xw + w'Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = x_0 = 0$,

$$0 = \int_0^\infty [x'\left(A'X + XA + F'B'X + XBF\right)x + x'Xw + w'Xx] dt.$$

Adding the last equation to the definition of J and completing squares

$$J = \int_0^\infty \{x'\left[A'X + XA + C'C - X\left(BB' - \gamma^{-2}I\right)X\right]x + \|Zx\|^2 - \|v\|^2\} dt$$

where $Z = F + B'X$ and $v = \gamma w - \gamma^{-1} X x$.

bii) The requirement $\|H\|_\infty < \gamma$ is equivalent to $J < 0$ when $x_0 = 0$. Thus sufficient conditions for $J < 0$ are $Z = 0$ and the existence of X such that

$$A'X + XA + C'C - X\left(BB' - \gamma^{-2}I\right)X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2} X$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_c x$ where $A_c = A - \left(BB' - \gamma^{-2}I\right)$ and a further condition is $Re\ \lambda_i(A_c) < 0,\ \forall\ i$.